Qwen3.8-27B + AgentDojo on Zaratan: setup

CMSC848N · Research Group 05 · from Ishaan · Oct 21, 2026, updated Oct 29. One script self-hosts Qwen3.8-27B with vLLM and runs the AgentDojo prompt-injection suite against it. Every version is pinned, so we all get the same numbers.

STATUS (THU OCT 29)

- **Validated end to end** on one A100: the smoke test, the Banking baselines (thinking low and off), the tool filter, delimiting and repeat-prompt defenses, and the custom attacks (`./make.sh attack`). One run takes 14 min to 1 h 23 min.
- **Read before you queue:** every run so far is in the shared `runs/` folder. Check there first so nobody spends SU on a duplicate.
- **Not in the image yet:** the ProtectAI detector defense (`transformers_pi_detector`) needs an image rebuild.

Class allocation: 33k of 100k SU left, shared with the other groups.

STEPS

1. **Check access.** Log in with your UMD password + Duo, then check that `cmsc848n-class` is listed:

   ```
   ssh <directory-id>@login.zaratan.umd.edu
   sbalance
   ```

2. **Get the scripts** (on Zaratan; the scripts live in `cluster/`):

   ```
   cd ~/scratch
   git clone -b cluster \
   https://github.com/qwillqwack/cmsc848n_project.git
   cd cmsc848n_project/cluster
   ```

3. **Setup** (on the login node). It finds the group's shared image and weights and returns in seconds:

   ```
   ./make.sh setup
   ```

4. **Smoke test:** one GPU job, up to 30 min (≤24 SU on an A100). Banking: pay a bill, total March spending, then the bill payment again with an injection hidden in the bill. Checks that tool calling works end to end.

   ```
   ./make.sh smoke
   ```

5. **Baseline:** the **banking suite**, our focus. Its 16 user tasks run with no attack (benign utility), then each against all 9 injection tasks with the `important_instructions` attack (144 runs), split across 4 parallel AgentDojo processes. Up to 2 h (≤96 SU); rerun with the same `RUN_DIR` to resume. `SUITES=all` adds workspace, slack and travel (up to 6 h).

   ```
   ./make.sh baseline
   ```

6. **Check on it:**

   ```
   ./make.sh status     # queue + newest logs
   ./make.sh results    # utility + ASR per suite
   ```

   Attack success rate (ASR) = share of attacked runs where the agent did what the injection asked; lower is better. AgentDojo calls it "security".

Everyone logs in with their own UMD account (any login node works; files are on shared scratch). The image, weights and results are shared, so **agree on one person per GPU run** and read the others' results in `runs/`. Duplicate jobs spend the class allocation and lengthen the queue.

SETTINGS

Set them before the command, e.g. `THINKING=off ./make.sh baseline`

`SUITES`	`banking` (default) or `all`
`THINKING`	Qwen3.8 reasoning effort: `off`, `low` (default), `medium`, `xhigh`.
`MAX_MODEL_LEN`	context length; default 32768. Lower it if vLLM runs out of memory.
`RUN_DIR`	reuse a run folder; finished tasks are skipped
`TIME_LIMIT`	Slurm walltime, e.g. `02:00:00`
`PROFILE=h100`	one H100 instead of an A100. Only with TA approval.

WHERE THINGS LIVE

`/scratch/zt1/project/cmsc848n/shared/proj05_agentdojo/`

- `images/`: the Apptainer image · `hf/`: model weights
- `runs/<time>-<mode>-a100-think-low/`: `results.txt`, `config.env` (exact settings), `vllm.log`, one log per shard, AgentDojo traces
- `runs/slurm/`: Slurm job logs

WHAT'S PINNED

vLLM 0.32.0 (image digest) · `Qwen/Qwen3.8-27B-FP8 @ 017b9c7` · AgentDojo 0.1.35 + every dependency hash-locked · benchmark v1.2.2

COURSE RULES (TA CANVAS)

- Charge jobs to `cmsc848n-class` and use one A100 (`make.sh` does both)
- No H100s or multi-GPU without a reason; ask the TAs about big jobs
- No heavy work on login nodes; compute nodes have no Internet
- Use the shared weights; don't download your own copy

IF SOMETHING BREAKS

- vLLM won't start: check the end of `vllm.log` in the run folder
- Out of GPU memory: `MAX_MODEL_LEN=16384 ./make.sh smoke`
- Off Zaratan: Docker + a ≥40 GB NVIDIA GPU, then `./make.sh all`